2 (a) e.g. time of collisions negligible compared to time between collisions

no intermolecular forces (except during collisions)

random motion (of molecules)

large numbers of molecules

(total) volume of molecules negligible compared to volume of containing vessel
or
average/mean separation large compared with size of molecules

any two B2 [2]

2 (b) (i) mass = $4.0 / (6.02 \times 10^{23}) = 6.6 \times 10^{-24}$ g
or
mass = $4.0 \times 1.66 \times 10^{-27} \times 10^{3} = 6.6 \times 10^{-24}$ g B1 [1]

(ii) $\frac{3}{2} kT = \frac{1}{2} m <c^2>$ C1

$\frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = \frac{1}{2} \times 6.6 \times 10^{-27} \times <c^2>$

$<c^2> = 1.88 \times 10^{6}$ ($m^2 s^{-2}$) C1

r.m.s. speed = 1.4×10^{3} $m s^{-1}$ A1 [3]